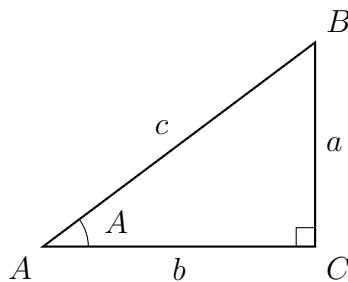


Right-Triangle Trigonometry — Wrong Ratio, Wrong Inverse, Wrong Mode

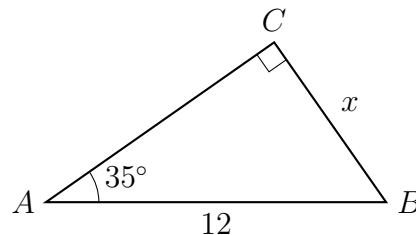
Three errors account for most lost marks in this unit: choosing the ratio that does not contain the two sides you have, putting the sides into an inverse function upside down, and leaving the calculator in radian mode.

- Before computing, name the two sides involved (opposite / adjacent / hypotenuse) and write the ratio you chose.
- Round side lengths and angle measures to two decimal places.
- Sanity-check every answer: a leg is always shorter than the hypotenuse, and an acute angle is between 0° and 90° .



How every triangle here is labelled. No values shown: use the numbers given in each problem.

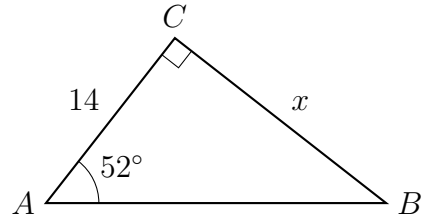
- 1.** The hypotenuse is 12 and $\angle A = 35^\circ$. Find x , the side opposite $\angle A$. Write the ratio you chose before you compute.



Answer: _____

2. Find and fix. Dana needs x , the side opposite $\angle A$, from the adjacent leg 14 and $\angle A = 52^\circ$. She writes the correct setup $x = 14 \tan 52^\circ$, presses enter, and her calculator returns -84.75 .

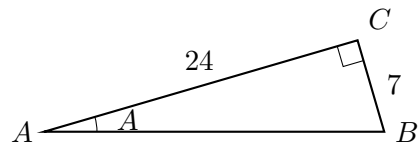
Her setup is right, so the fault is in the calculator. Say what it is, then give the correct value of x .



Answer: _____

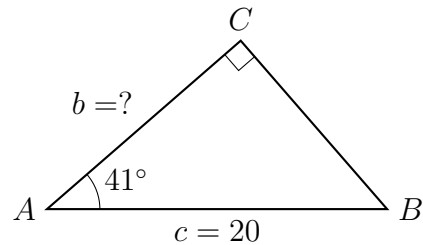
3. Find and fix. Eli must find $\angle A$ from the two legs, 7 opposite $\angle A$ and 24 adjacent to it. He writes $\angle A = \tan^{-1}\left(\frac{24}{7}\right) = 73.74^\circ$.

His arithmetic is fine; the ratio inside the inverse is not. Name the error, then give $\angle A$ correctly.



Answer: _____

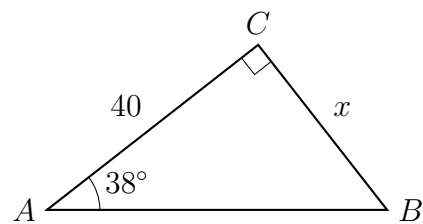
4. The hypotenuse is 20 and $\angle A = 41^\circ$. Find b , the leg *adjacent* to $\angle A$. A classmate reaches for sine out of habit; decide whether that is the ratio this side needs, then compute.



Answer: _____

5. **Find and fix.** Ravi stands 40 feet from the base of a flagpole on level ground. The angle of elevation from his eye level to the top of the pole is 38° . He sets up the height as $40 \tan 38^\circ$ and his screen shows 12.41.

That height is less than half the distance he is standing away, even though the angle is close to 45° — so the number cannot be right. Find what his calculator is doing and give the correct height of the pole above his eye level, in feet.

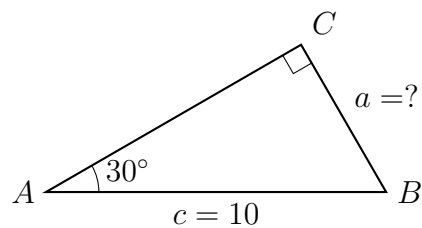


Answer: _____ ft

6. This is a 30-60-90 triangle: the hypotenuse is 10 and $\angle A = 30^\circ$.

(a) Find a , the side opposite the 30° angle.

(b) Sasha answers 8.66 for part (a). Which side of this triangle is 8.66, and which ratio did Sasha use to get it? Answer in one sentence beside your work.



Answer: _____